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Exact solutions to robust control problems involving scalar hyperbolic conservation laws using Mixed Integer Linear Programming

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Abstract— **This article presents a new robust control framework for transportation problems in which the state is modeled by a first order scalar conservation law. Using an equivalent formulation based on a Hamilton-Jacobi equation, we pose the problem of controlling the state of the system on a network link, using boundary flow control, as a Linear Program. Unlike many previously investigated transportation control schemes, this method yields a globally optimal solution and is capable of handling shocks (*i.e.* discontinuities in the state of the system). We also demonstrate that the same framework can handle robust control problems, in which the uncontrollable components of the initial and boundary conditions are encoded in intervals on the right hand side of inequalities in the linear program. The lower bound of the interval which defines the smallest feasible solution set is used to solve the robust LP (or MILP if the objective function depends on boolean variables). Since this framework leverages the intrinsic properties of the Hamilton-Jacobi equation used to model the state of the system, it is extremely fast. Several examples are given to demonstrate the performance of the robust control solution and the trade-off between the robustness and the optimality.**

I. Introduction

Physical systems driven by *Partial Differential Equations* (PDEs) are very common in engineering, biology, physics and chemistry. Such systems are often referred to as *Distributed Parameter Systems*, as their state is infinite dimensional. In the context of transportation engineering, the flow of traffic is usually modeled as such, using various classes of PDEs, most notably first [1], [2] and second order [3], [4], [5] traffic flow models.

Controlling transportation networks (which can be modeled by the previously cited PDEs) is of critical importance, and could have a dramatic economic and societal impact. Traffic congestion is indeed a considerable issue worldwide and is expected to become worse as global traffic demand increases. The current cost of congestion is estimated at \$700 per driver in the USA, more than one week of wages for the average US citizen. Furthermore, traffic congestion causes an extra 1.9 billion gallons of fuel being wasted in the USA alone, about six days of daily fuel usage for ground transportation.

Reducing traffic congestion can be achieved by increasing the capacity of roads (which is expensive), modifying user demand, or actively controlling traffic using for instance adaptive speed limits, ramp metering, or traffic signal control. Traffic flow control [6], [7], [8] is a very promising direction for reducing traffic congestion, as it does not require expensive road construction, and does not require the control of user demand, which is sometimes impractical (for instance during work hours). However, to date, traffic control is limited to a few small scale experiments, mainly due to the lack of accurate traffic data and the lack of computationally efficient optimal control methods.

While a large number of optimal control methods exist for *Lumped Parameter Systems*, relatively few methods exist for the stabilization and control of systems modeled by PDEs. A large number of estimation and control methods for PDEs rely on model reduction, that is, the conversion of the PDE into a discrete or continuous time lumped parameter system [9]. Examples of such methods include [10], [11], [12], [13], [14], [15], [16]. Some methods however do not require such an approximation, such as backstepping methods in [17] for chemical reactors, Lyapunov methods in [18] for the the stabilization of the Burgers equation, and Lyapunov stability analysis of equilibria in networks of scalar conservation laws [19]. To the best of our knowledge, [18] is the first successful application of a Lyapunov method to stabilize an hyperbolic PDE with shocks, which is related to the well known *Lighthill Whitham Richards* (LWR) traffic flow model. In the context of transportation network control, we are not only interested in stabilization, but also in control, to minimize certain objectives, for instance the total travel time, worst case travel time, minimal fuel consumption. Methods for the optimal control on large scale networks [20], [21] and with source terms[22] have also been discussed. However, in the presence of the environmental noise and measurement error, the optimal control output could likely to be unpractical in applications. The measurement uncertainty should be taken into account while discussing the optimal solution which hence converges to a robust control solution.

In this article, we propose a new method for controlling the LWR PDE with triangular flux function on transportation networks. Using an equivalent *Hamilton Jacobi* formulation of the LWR PDE, we show that the problem of controlling the traffic density on a highway link by modifying its boundary conditions (boundary control) can be posed as a *Linear Program* (LP), or as a *Mixed Integer Linear Program* (MILP) depending on the formulation of the objective function to minimize. To the best of our knowledge, this is the first computational method that leverages the intrinsic properties of the LWR equations to control them (or more generally first order scalar conservation laws with piecewise linear

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flux functions). In particular, the proposed method yields a significantly reduced complexity compared to standard optimization methods for lumped parameter systems. We further extend this framework to solve robust control problems in which some uncontrollable components of the solution are not precisely known.

The rest of this article is organized as follows. Section II reviews the framework for the traffic estimation developed in [23] using the LWR traffic flow model. We then derive the optimal control problem as a LP (or a MILP depending on the objective function) using the traffic density estimation investigated earlier. Section III presents the traffic control problem, as well as a toy boundary traffic control example on a single highway link. We then introduce robust control problems with data uncertainty encoded in the right hand side intervals in LP inequality constraints in Section IV.

II. Model Definition

A. Traffic flow model

In this article, we define one link on a highway section as $P := [\xi, \chi]$ where ξ and χ respectively represent the upstream and downstream post miles of the link on the highway. On this highway link, traffic can be macroscopically described using the density function $\rho(t,x)$, which evolves according to a traffic model. One of the most commonly used macroscopic models for the traffic flow is the *Lighthill-Whitham-Richards* (LWR) model [1], [2].

$$\frac{\partial \rho(t,x)}{\partial t} + \frac{\partial \psi(\rho(t,x))}{\partial x} = 0 \tag{1}$$

Alternatively, the state of traffic can also be described as a time and space dependent scalar function $\mathbf{M}(\cdot,\cdot)$, known as the *Moskowitz function* [24], [25], which is obtained by integration of the density function. The Moskowitz function satisfies the following *Hamilton-Jacobi*(HJ) PDE, obtained by integration of the LWR PDE:

$$\frac{\partial \mathbf{M}(t,x)}{\partial t} - \psi\left(-\frac{\partial \mathbf{M}(t,x)}{\partial x}\right) \;=\; 0 \tag{2}$$

In the above equation, the function $\psi(\cdot)$ is known as the *Hamiltonian*. In the remainder of this article, the *Hamiltonian* is assumed to be the following continuous and concave triangular function:

$$\psi(\rho) = \begin{cases} v_f \rho & : \rho \in [0, \rho_c] \\ w(\rho - \rho_\kappa) & : \rho \in [\rho_c, \rho_\kappa] \end{cases} \tag{3}$$

where ρ_c, ρ_κ, v_f and w are parameters representing the critical density (ρ_c), the maximal density (ρ_κ), the free flow speed (v_f) and the congestion speed (w) and satisfying in the following formula. In this article, these parameters are assumed to be static, obtained from the construction standards or estimated as in [26].

$$\rho_c = \frac{-w\rho_\kappa}{v_f - w}$$

While the method investigated in this article could be applied to general concave flux functions, a piecewise linear flux function allows us to pose the control problem as a *Mixed Integer Linear Program* (MILP), while general concave flux functions would result in a *Mixed Integer Convex Program*, for which no efficient computational methods are currently available. Note that triangular fundamental diagrams are commonly used in traffic [27], [28], [29], since they are very robust and depend on a minimum number of parameters that are easy to physically interpret.

B. Solutions to the Lighthill-Whitham-Richards equation

One of the several classes of weak solutions to equation (2) is the *Barron-Jensen/Frankowska* (B-J/F) solution [30], [31], which are investigated in this article. The B-J/F solutions to equation (2) are fully characterized by the *Lax-Hopf* formula, which was initially derived using the control framework of viability theory. Note that B-J/F solutions to Hamilton Jacobi PDEs are equivalent to viscosity solutions [32] in the present case, since no internal conditions [33] are considered in this article.

Solving partial differential equations requires the definition of initial and boundary conditions, which can be encoded are as the general concept of value conditions [34].

*Definition 2.1 (**Value Condition**):* A value condition $\mathbf{c}(\cdot,\cdot)$ is a lower semi-continuous function defined on a subset of $[0, t_{max}] \times [\xi, \chi]$.

In this article, we assume the value conditions are affine functions of time and space, defined on a line segment of $\mathbb{R}_+ \times X$. Given an arbitrary value condition $\mathbf{c}(\cdot,\cdot)$, we define its associated solution $\mathbf{M_c}(\cdot,\cdot)$ to HJ PDE (2) using the following Lax-Hopf formula [35].

*Proposition 1 (**Lax-Hopf formula**):* Let $\psi(\cdot)$ be a concave and continuous Hamiltonian, and let $\mathbf{c}(\cdot,\cdot)$ be a value condition, as in Definition 2.1. The B-J/F solution $\mathbf{M_c}(\cdot,\cdot)$ to (2) associated with $\mathbf{c}(\cdot,\cdot)$ is defined algebraically by:

$$\mathbf{M_c}(t,x) = \inf_{(u,T)\in \mathrm{Dom}(\varphi^*)\times\mathbb{R}_+} \left(\mathbf{c}(t-T, x+Tu) + T\varphi^*(u)\right) \tag{4}$$

where the function $\varphi^*(\cdot)$ is the Legendre-Fenchel transform of the upper semicontinuous Hamiltonian $\psi(\cdot)$, given by:

$$\varphi^*(u) := \sup_{p\in \mathrm{Dom}(\psi)} [p \cdot u + \psi(p)]$$

The structure of the Lax-Hopf formula (4) implies the following important property, known as *inf-morphism* property [35].

*Proposition 2 (**Inf-morphism property**):* Let the value condition $\mathbf{c}(\cdot,\cdot)$ be minimum of a finite number of lower semicontinuous functions:

$$\forall (t,x) \in [0, t_{\max}] \times [\xi, \chi], \;\; \mathbf{c}(t,x) := \min_{j\in J} \mathbf{c}_j(t,x)$$

The solution $\mathbf{M_c}(\cdot,\cdot)$ associated with the above value condition can be decomposed [35], [36], [33] as:

$$\forall (t,x) \in [0, t_{\max}] \times [\xi, \chi], \;\; \mathbf{M_c}(t,x) = \min_{j\in J} \mathbf{M}_{\mathbf{c}_j}(t,x)$$

Because of the *Inf-morphism property*, we can instantiate the LWR PDE model constraints as a set of inequalities.

*Proposition 3 (**Compatibility conditions**):* Let the value condition $\mathbf{c}(\cdot,\cdot) = \min_{j\in J} \mathbf{c}_j(t,x)$, and the associated solution $\mathbf{M_c}(\cdot,\cdot)$ defined in (4). The equality $\forall (t,x) \in \mathrm{Dom}(\mathbf{c})$,

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$\mathbf{M_c}(t,x) = \mathbf{c}(t,x)$ is valid if and only if the following inequality constraints are satisfied:

$$\mathbf{M}_{\mathbf{c}_j}(t,x) \geq \mathbf{c}_i(t,x) \quad \forall (t,x) \in \mathrm{Dom}(\mathbf{c}_i), \forall (i,j) \in J^2$$

The proof of this proposition is available in [34]. In this article, we only consider initial, upstream and downstream boundary conditions. For simplicity, we assume that these conditions are piecewise linear, and continuous, which can be physically interpreted as piecewise constant initial densities and piecewise constant boundary flows.

Definition 2.2: Let us define $\mathbb{K}= \{0,\dots,k_{max}\}$, and $\mathbb{N} = \{0,\dots,n_{max}\}$. For all $k \in \mathbb{K}$, and $n \in \mathbb{N}$, we define the following functions, respectively called initial, upstream and downstream (boundary) conditions:

$$M_k(t,x)=\begin{cases} -\sum_{i=0}^{k-1}\rho(i)X \\ -\rho(k)(x-kX) & \text{if } t=0 \\ & \text{and } x \in [kX,(k+1)X] \\ +\infty & \text{otherwise} \end{cases}$$

$$\gamma_n(t,x)=\begin{cases} \sum_{i=0}^{n-1} q_{\text{in}}(i)T \\ +q_{\text{in}}(n)(t-nT) & \text{if } x=\xi \\ & \text{and } t \in [nT,(n+1)T] \\ +\infty & \text{otherwise} \end{cases}$$

$$\beta_n(t,x)=\begin{cases} \sum_{i=0}^{n-1} q_{\text{out}}(i)T \\ +q_{\text{out}}(n)(t-nT) \\ -\sum_{k=0}^{k_{max}}\rho(k)X & \text{if } x=\chi \\ & \text{and } t \in [nT,(n+1)T] \\ +\infty & \text{otherwise} \end{cases}$$

where $\rho(i)$, $q_{in}(i)$, and $q_{out}(i)$ are the constant density, upstream and downstream flux value in their respective space and time intervals.

Given the initial, upstream and downstream value conditions, the corresponding solutions defined by the *Lax-Hopf* formula (4) can be explicitly expressed [34], [37] as follows:

$$\mathbf{M}_{M_k}(t,x) = \begin{cases} +\infty & \text{if } x \leq kX + wt \\ & \text{or } x \geq (k+1)X + vt \\ -\sum_{i=0}^{k-1}\rho(i)X \\ +\rho(k)(tv+kX-x) & \text{if } kX+tv \leq x \\ \qquad \text{and} & (k+1)X+tv \geq x \\ \qquad \text{and} & \rho(k) \leq \rho_c \\ -\sum_{i=0}^{k-1}\rho(i)X \\ +\rho_c(tv+kX-x) & \text{if } kX+tv \geq x \\ \qquad \text{and} & kX+tw \leq x \\ \qquad \text{and} & \rho(k) \leq \rho_c \\ -\sum_{i=0}^{k-1}\rho(i)X \\ +\rho(k)(tw+kX-x) \\ -\rho_m tw & \text{if } kX+tw \leq x \\ \qquad \text{and} & (k+1)X+tw \geq x \\ \qquad \text{and} & \rho(k) \geq \rho_c \\ -\sum_{i=0}^{k}\rho(i)X \\ \rho_c(tw+(k+1)X-x) \\ -\rho_m tw & \text{if } (k+1)X+tv \geq x \\ \qquad \text{and} & (k+1)X+tw \leq x \\ \qquad \text{and} & \rho(k) \geq \rho_c \end{cases}$$

$$\mathbf{M}_{\gamma_n}(t,x) = \begin{cases} +\infty & \text{if } t \leq nT + \frac{x-\xi}{v} \\ \sum_{i=0}^{n-1} q_{\text{in}}(i)T \\ +q_{\text{in}}(n)(t-\frac{x-\xi}{v}-nT) & \text{if } nT+\frac{x-\xi}{v} \leq t \\ & \text{and } t \leq (n+1)T \\ & +\frac{x-\xi}{v} \\ \sum_{i=0}^{n} q_{\text{in}}(i)T \\ +\rho_c v(t-(n+1)T-\frac{x-\xi}{v}) & \text{otherwise} \end{cases}$$

$$\mathbf{M}_{\beta_n}(t,x) = \begin{cases} +\infty & \text{if } t \leq nT \\ & +\frac{x-\chi}{w} \\ -\sum_{k=0}^{k_{max}}\rho(k)X + \sum_{i=0}^{n-1} q_{\text{out}}(i)T \\ +q_{\text{out}}(n)(t-\frac{x-\chi}{w}-nT) \\ -\rho_m(x-\chi) & \text{if } nT \\ & +\frac{x-\chi}{w} \leq t \\ \qquad \text{and} & t \leq (n+1)T \\ & +\frac{x-\chi}{w} \\ -\sum_{k=0}^{k_{max}}\rho(k)X + \sum_{i=0}^{n} q_{\text{out}}(i)T \\ +\rho_c v(t-(n+1)T-\frac{x-\chi}{v}) & \text{otherwise} \end{cases}$$

The explicit solutions enable us to write down the constraints encoded by the LWR PDE model (1) as a set of linear inequalities [34], using the *inf-morphism property* and the *compatibility conditions*. Therefore, model constraints for the solutions associated to the initial, upstream and downstream conditions can be defined as following inequalities [23] [38].

$$\begin{cases} \mathbf{M}_{M_k}(0,x_p) \geq M_p(0,x_p) & \forall (k,p) \in \mathbb{K}^2 \\ \mathbf{M}_{M_k}(pT,\chi) \geq \beta_p(pT,\chi) & \forall k \in \mathbb{K}, \forall p \in \mathbb{N} \\ \mathbf{M}_{M_k}(\frac{\chi-x_{k+1}}{v},\chi) \geq \\ \beta_p(\frac{\chi-x_{k+1}}{v},\chi) & \forall k \in \mathbb{K}, \forall p \in \mathbb{N} \text{ s. t.} \\ & \frac{\chi-x_{k+1}}{v} \in [pT, \\ & (p+1)T] \\ \mathbf{M}_{M_k}(pT,\xi) \geq \gamma_p(pT,\xi) & \forall k \in \mathbb{K}, \forall p \in \mathbb{N} \\ \mathbf{M}_{M_k}(\frac{\xi-x_k}{w},\xi) \geq \gamma_p(\frac{\xi-x_k}{w},\xi) & \forall k \in \mathbb{K}, \forall p \in \mathbb{N} \text{ s. t.} \\ & \frac{\xi-x_k}{w} \in [pT,(p+1)T] \end{cases}$$

$$\begin{cases} \mathbf{M}_{\gamma_n}(pT,\xi) \geq \gamma_p(pT,\xi) & \forall (n,p) \in \mathbb{N}^2 \\ \mathbf{M}_{\gamma_n}(pT,\chi) \geq \beta_p(pT,\chi) & \forall (n,p) \in \mathbb{N}^2 \\ \mathbf{M}_{\gamma_n}(nT+\frac{\chi-\xi}{v},\chi) \geq \beta_p(nT+\frac{\chi-\xi}{v},\chi) & \forall (n,p) \in \mathbb{N}^2 \text{ s. t. } nT+ \\ & \frac{\chi-\xi}{v} \in [pT,(p+1)T] \end{cases}$$

$$\begin{cases} \mathbf{M}_{\beta_n}(pT,\xi) \geq \gamma_p(pT,\xi) & \forall (n,p) \in \mathbb{N}^2 \\ \mathbf{M}_{\beta_n}(nT+\frac{\xi-\chi}{w},\xi) \geq \gamma_p(nT+\frac{\xi-\chi}{w},\xi) & \forall (n,p) \in \mathbb{N}^2 \text{ s. t. } nT+ \\ & \frac{\xi-\chi}{w} \in [pT,(p+1)T] \\ \mathbf{M}_{\beta_n}(pT,\chi) \geq \beta_p(pT,\chi) & \forall (n,p) \in \mathbb{N}^2 \end{cases}$$

An important property of these constraints is that they are all linear in the unknown initial and boundary condition coefficients $(\rho(\cdot), q_{in}(\cdot), q_{out}(\cdot))$. Therefore, we can write these inequalities in a more concise matrix form:

$$A_{model} y \leq b_{model}$$

where y is a column vector variable, consisting of initial and boundary condition coefficients $\rho(\cdot), q_{in}(\cdot), q_{out}(\cdot)$. A_{model} and b_{model} are respectively the coefficients of the variable y and the right hand constants in the above inequalities.

While these initial and boundary condition coefficients are subject to the above model constraints, measurement

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data yields additional constraints on their possible values. However, physically it is impossible for measurements to perfectly represent the real values. One simple approach in practice is using the nominal value without considering the measurements error, assuming that the measurement error is not sufficiently large to cause significant impact to the optimal solution. While in problems where the measurements data plays a critical role, the uncertainty must be integrated into the model.

Multiple error models are possible, and can lead to different optimization problems. One of the simplest error model is to assume bounds on the values of the densities and flows associated to the uncontrolled components of the initial and boundary conditions. This error model yields box constraints, which could turn the robust control problem into an interval linear programming with inequality constraints (provided that the objective is linear). Another error model is considering interval values as random quantities with certain distribution which boils to stochastic programming [39]. While the box constraints model is simple and practical in describing data uncertainty, we assume that the uncertainty is enclosed by bounds on the densities and flow constraints.

As an example, if we consider sensors measuring the boundary flows $(q_{in/out}(0,\cdot), \cdots q_{in/out}(n_m,\cdot))$ and the initial density $(\rho(0,0), \cdots \rho(0,k_m))$ with a relative uncertainty of $r_q\%$ and $r_\rho\%$ respectively, the corresponding measurement data constraints are:

$$\begin{cases} (1-r_q/100)q^{\text{meas}}_{\text{in/out}}(n) \leq q_{\text{in/out}}(n) \leq (1+r_q/100)q^{\text{meas}}_{\text{in/out}}(n) \\ \qquad \forall n \in [0, n_{\max}] \\ (1-r_\rho/100)\rho(k)^{\text{meas}} \leq \rho(k) \leq (1+r_\rho/100)\rho(k)^{\text{meas}} \\ \qquad \forall k \in [0, k_{\max}] \end{cases}$$

Thus, defining y as the decision variable: $y := \big(\rho_{ini}(1), \ldots, \rho_{ini}(k_{max}), q_{in}(1), \ldots, \quad q_{in}(n_m), q_{out}(1), \ldots q_{out}(n_{max})\big)$, a traffic estimation problem can be posed as the following optimization program, with linear constraints:

$$\begin{array}{l} \text{Minimize } \ f(y) \\ \text{s. t. } \begin{cases} A_{model}y \leq b_{model} \\ A_{data}y = b_{data} \in \big[\underline{b}, \overline{b}\big] \end{cases} \end{array} \tag{5}$$

In the above program, $f(\cdot)$ is the objective function (used to select a solution among all solutions satisfying the model and measurement data constraints) for the traffic estimation under the compatibility constraints from the physical conditions and sensor measurements. $\big[\underline{b}, \overline{b}\big]$ are respectively element-wise lower and upper bounds for b_{data}. See [40] for applications of this framework to various traffic flow estimation problems.

III. Traffic control on a single highway link

In the previous section, we showed that traffic flow estimation problems could be formulated as optimization programs with linear constraints. In this section, we briefly extend this formulation to traffic flow control problems assuming that all the measurement data are accurate $\underline{b} = \overline{b}$.

A. Linear Constraints

Model constraints originate from the physics of traffic flow propagation, and thus remain identical in this control application. Measurement data constraints A_{data}, b_{data} in LP (5) also remain identical, though for the sake of boundary control we have to remove a subset of these constraints to simulate a given control input.

In this example, we assume that initial conditions are given by setting prescribed measurements $\rho(m)^{meas}$ and cannot be controlled. While the data constraints for the boundary flows $q_{in/out}$ are relaxed for all time thus boundary flows can be regarded as our control input to the system. Note that our framework is also capable of only controlling any arbitrary subset of the boundary flow variables $q_{in/out}(t)$.

B. Objective function

Several objective functions can be considered when solving the optimal control problem. Possible functions include the total amount of downstream flow in given simulation time t_{max}, the total time needed to clear congestion or the least amount of congestion at the final time.

In this example, we choose for instance to maximize downstream flow, leading to the following linear program:

$$\begin{array}{l} \text{Minimize } \ -\sum_{i=1}^{n_{max}} q_{out}(i) \\ \text{s. t. } \begin{cases} A_{model}y \leq b_{model} \\ \rho(k) = \rho(k)^{\text{meas}} \end{cases} \end{array} \tag{6}$$

C. Implementation

In this implementation, we consider a spatial domain of 3.858 km, located between the PeMS VDSs (vehicle detection stations) 400536 and 400284 on Highway I-880 N around Hayward, California.

The link is divided into 6 segments of equal length $X = 643\,m$. The initial density measurements on 6 segments are defined as 6 piece-wise affine constants in the range in the range $[0.5\rho_c, 3\rho_c]$ that represent a partly congested link of a highway. We simulate 15 boundary conditions with a granularity of T= 30 s.

We solve the LP (6) using `IBM Ilog Cplex 12.4` on a `Macbook` operating `MacOS X`. As illustrated in Fig. 1, the optimal control of the upstream and downstream flows alleviates traffic congestion by limiting its upstream incoming flow and eventually achieves a free flow condition in less than 7 minutes (simulation time). This particular optimization problem involves 36 variables and 600 constraints, and is solved in 0.02 s.

IV. Robust control problem formulation

Since experimental traffic data is usually very scarce in practice, traffic estimates can be highly uncertain. Thus, assuming that the initial densities are perfectly known may lead to a dramatic decrease in the performance of the optimal control in practice [41]. To solve this issue, we are interested in the *robust control* problem, in which the uncertainty from measurement data is modeled as intervals and encoded in the right hand side in the linear program.

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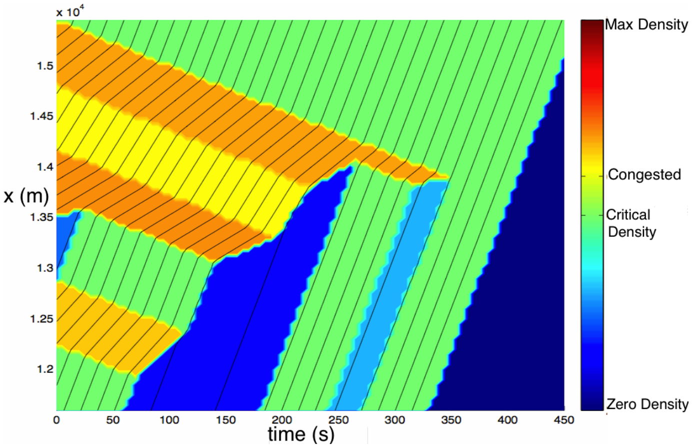

Fig. 1. **Boundary control on a single link**: In this example, we show the optimal boundary control problem. The initial condition is assumed to be 6 piecewise constants and partially congested. We control the upstream and downstream boundary flows to maximize the cumulative downstream flow in the interval $[0, t_{max}]$. As we can see, the congestion propagates backward on the road based on the traffic physics. And the downstream flows are all the maximal allowable flow during the simulation time which certainly is an optimal solution to our objective function. Also please note that the last portion of the upstream flows are zero since they do not contribute to our objective in this example.

A. Problem definition

In this section, we are interested as previously in the boundary control problem. However we now assume that the uncontrolled segments of the initial or boundary conditions can exhibit uncertainty.

We consider the following standard form of the linear program defined previously:

$$\begin{array}{ll} \text{Minimize} & f(y) \\ \text{s. t.} & A_{model} y \leq b_{model} \\ & A_{data} y = b \end{array} \tag{7}$$

We define that each b_i could take any value in the interval $\left[\underline{b_i}, \overline{b_i}\right]$ where $\underline{b_i} = b_i^{meas} \cdot (1 - r_i)$ and $\overline{b_i} = b_i^{meas} \cdot (1 + r_i)$. We also assume each b_i varies in the interval regardless of values taken by other coefficients. Each set of $b \in \left[\underline{b}, \overline{b}\right]$ (lower and upper bounds are defined component-wise) represents one linear program scenario in the optimal control problem. The constants r_i represent the allowed relative error on b_i.

Our objective is to find the robust solution that is feasible for all sets of $b \in \left[\underline{b}, \overline{b}\right]$ which realistically represents the optimal control outputs that are compatible with the physical limitations and the data uncertainty.

B. Robust MILP formulation

In the robust control problem, we separate the controllable variables y_c with data-constrained variables y_d in the previous decision variable $y := [y_d; y_c]$.

Therefore, the inequality for model constraints can be reformulated as following:

$$[A_d \;\; A_c] \begin{bmatrix} y_d \\ y_c \end{bmatrix} \leq b_{model}$$

which is equivalent to

$$A_c \, y_c \leq b_{model} - A_d \, y_d \tag{8}$$

In our framework, densities and boundary flows are a linear function of the decision variable. Thus, the equality in the LP (7) for data constraints $A_{data} \cdot y = b$ could be expressed more directly as

$$y_d = y^{meas} \in [\underline{y}^{meas} \;\; \overline{y}^{meas}]$$

where $\underline{y}^{meas}$ and $\overline{y}^{meas}$ are the element-wise lower and upper bounds.

If we replace y_d by the measurements y^{meas} in Equation (8), the right hand side becomes

$$b_c = b_{model} - A_d y^{meas}$$

We thus define the lower and upper bounds for b_c respectively as

$$\underline{b}_c := [\underline{b}_{c\,1}, \ldots \underline{b}_{c\,i}, \ldots]^T$$

where

$$\underline{b}_{c\,i} = \{\min(b_{model}(i) - A_d(i,:)\, y^{meas}) \mid \text{s.t. } \underline{y}^{meas} \leq y^{meas} \leq \overline{y}^{meas}\}$$

and $\overline{b}_c$ respectively.

Note that in practice, $\underline{b}_c$ and $\overline{b}_c$ are not necessarily reachable since the lower/upper bound of each element $b_{c\,i}$ might correspond to a different set of measurement data.

By encoding the data uncertainty as the right hand side interval $[\underline{b}_c \; \overline{b}_c]$, the LP (7) becomes:

$$(P^{b_c}) \left\{ \begin{array}{ll} \text{Minimize} & f(y_c) \\ \text{s. t.} & A_c y_c \leq b_c \\ & \underline{b}_c \leq b_c \leq \overline{b}_c \end{array} \right. \tag{9}$$

For $\underline{b}_c \leq b_c \leq \overline{b}_c$, we denote as Y^{b_c} the polyhedron defined by $\{y_c \in \mathbb{R}^n | \, A_c \, y_c \leq b_c\}$. This polyhedron represents the feasible set for the linear program (P^{b_c}).

Since $Y^{\underline{b}_c} \subseteq Y^{b_c}$, all solutions to linear program $(P^{\underline{b}_c})$ satisfy the inequality constraints in (P^{b_c}) regardless of the data uncertainty [42] [43]. Hence the robust control problem can be formulated as the following linear program:

$$(P^{\underline{b}_c}) \left\{ \begin{array}{ll} \text{Minimize} & f(y_c) \\ \text{s. t.} & A_c y_c \leq \underline{b}_c \end{array} \right. \tag{10}$$

C. Applications

In this section, we demonstrate the applicability of the method on a robust traffic control example inspired by the control problems investigated in Section III.

The link is divided into 10 segments with equal length $X = 300\,m$. The nominal measurement data for initial conditions are assumed to be $\rho^{meas} = [3, 5, 2, 8, 6, 9, 10, 7, 1, 4] \times 0.5\rho_c$ respectively for 10 segments with an uncertainty level 10%.

We assume that the up/downstream boundary flows can be perfectly controlled and are not constrained by measurement data, though partial control of the boundary conditions can be achieved (see Section III) and can be formulated as Model Predictive Control.

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Based on the formulation of the robust linear program in the previous subsection, the decision variable y_c only includes the controllable boundary flows $y_c := [q_{in}(1), \ldots, q_{in}(n_{max}), q_{out}(1), \ldots q_{out}(n_{max})]$, while ρ^{meas} is encoded to the bounds of the right hand side $[\underline{b}_c\ \bar{b}_c]$.

We pose the objective function as:

$$f(y_c) = -\sum_{i=1}^{n}\sum_{t=1}^{n_{max}} w(t) \cdot (q_{out}(i) + q_{in}(i)) + \sum_{i=1}^{30} b_{cong}(i)$$

where $w(t) = e^{5\cdot(t_{max}-t)/t_{max}}$ and b_{cong} is a binary variable. Setting $b_{cong}(i) = 1$ amounts to having the segment i congested at the time horizon t_{max}. The total upstream flows are added up in the objective function to obtain a more visible difference between the robust solution and the optimal solution, as shown later.

Due to the introduction of the congested segments which require additional binary variables in the decision variable y_c, the robust formulation problem becomes a robust mixed integer linear program which is obtained directly from the formulation (10).

We solve the robust mixed integer linear program (10) using `IBM Ilog Cplex 12.4` on a `Macbook` operating `MacOS X`. The MILP was solved in 0.04 s. The simulation results are illustrated in Figure 2. The figure plots the scenario in which all initial conditions are set by the nominal values without uncertainty. Clearly, to guarantee robustness for all possible data sets, the robust control outputs admit less upstream flows comparing to the optimal solution (in the same scenario).

We now consider two extreme scenarios in which the initial condition are alternatively set to their lower bounds or their upper bounds. We illustrate these scenarios in Figure 3, to further demonstrate the performance of the robust control algorithm. In these two scenarios, the robust control can still maintain a reasonable performance, though appearing too conservative in some cases, while the original optimal control solution (in the nominal value scenario) is infeasible in some of these cases.

Regarding the infeasibility of the optimal control solution in the situations where measurement error exists, let us consider one simple example. Assume that the nominal value of the initial density on the first segment of the link is $\rho(1)_{nom} = \rho_c$ and the relative measurement error is 10%, then the real initial density should be $\rho(1) \in [0.9\rho_c, 1.1\rho_c]$. The optimal controller takes the nominal value as the real initial condition without considering the measurement error and could result in a solution in which the upstream flow in the first time step $q_{in}(1) = q_{max}$. Note that $q_{in}(1) = q_{max}$ could occur only when $\rho(1) \leq \rho_c$ according to the *triangular fundamental diagram*. Therefore, if the real value of the initial condition $\rho(1) > \rho_c$ which is possible since $\rho(1) \in [0.9\rho_c, 1.1\rho_c]$, the optimal control outputs (where $q_{in}(1) = q_{max}$) would be infeasible in practice.

V. Conclusion

This article proposes a new robust control framework for transportation problems modeled by the Lighthill Whitham Richards partial differential equation. We first present an equivalent formulation of the problem based on a Hamilton Jacobi equation. The Lax-Hopf formula and Inf-morphism property of the solutions to the Hamilton Jacobi equation enable us to derive the constraints from the model as a set of linear inequalities in some decision variable. This enables us to pose the problem of single link control as an optimization program, most often a LP or a MILP (if we have boolean variables introduced by the objective function).

An example is presented to show the applicability of this framework by implementing continuous boundary flow control (for instance as ramp metering). The flexibility of this framework allows any subset of the variables of interest (densities and flows) to be constrained by experimental measurements while the remaining variables are control inputs. This could enable us to further develop this method into a model predictive control framework in which boundary control inputs are computed iteratively using feedback measurement data.

This framework can also be easily extended to handle robust traffic control problems. In this situation, the data uncertainty arising from sensor measurements is encoded as intervals on the right hand side of the inequalities in the LP. Therefore, solving a robust control problem boils down to solving a LP constrained by the lower bound of the interval which has the smallest feasible solution set.

To the best of our knowledge, this framework is the first to offer a computationally efficient solution method to the problem of robust control involving scalar hyperbolic conservation laws. It also has the advantage of being very efficient and flexible, that is, very different control problems can be solved using the same framework, since the decision variables can be all used as control variables or as fixed values (with possibly some uncertainty around these values).

Future work will involve developing the framework into a practical model predictive control framework leveraging its flexibility by using the fact that the boundary conditions can be partially controlled. Another important research direction is to incorporate the uncertainty of other coefficients into the model which is of critical importance for practical traffic flow control. Currently, the controllable value conditions are assumed to be perfectly controlled, while in a real system actuator noise would be present. With additional uncertainty integrated in the model, the robust solution might be too conservative to be applicable. Therefore, other algorithms that mildly sacrifice the robustness of the control outputs [44] [45] might provide more practical solutions which worth further investigation.

References

[1] M. Lighthill and G. Whitham, "On kinematic waves. ii. a theory of traffic flow on long crowded roads," *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, vol. 229, pp. 317–345, nov 1999.

[2] P. Richards, "Shock waves on the highway," *Operations research*, vol. 4, no. 1, pp. 42–51, 1956.

[3] A. Aw and M. Rascle, "Resurrection of second order models of traffic flow," *SIAM journal on applied mathematics*, vol. 60, no. 3, pp. 916–938, 2000.

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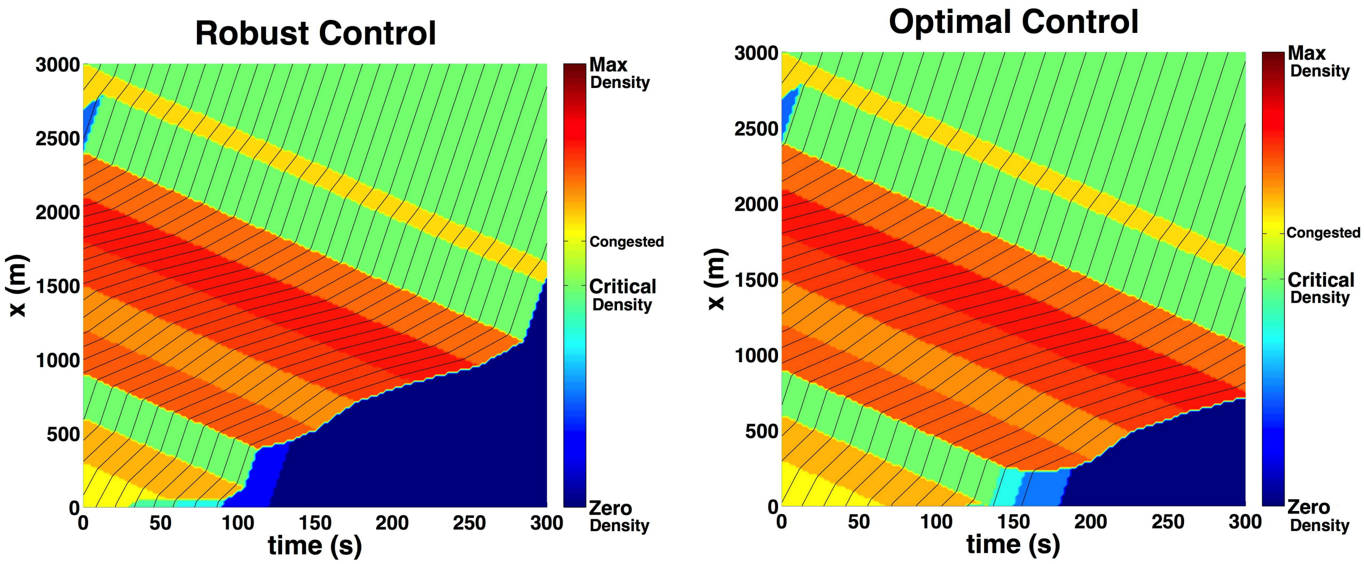

Fig. 2. **Comparison between optimal control and robust control of a highway segment:** In this Figure, we compare the solutions to the original problem and to the robust problem corresponding to the nominal values of the initial density. **Left:** This figure illustrates the robust control outputs associated with the MILP (10) with an uncertainty level 10% from the measurement data. **Right:** Optimal solution in the same scenario. From these Figures, we can see that the robust control solution is relatively conservative, admitting less upstream flows. It is thus less optimal than the solution to the original control problem.

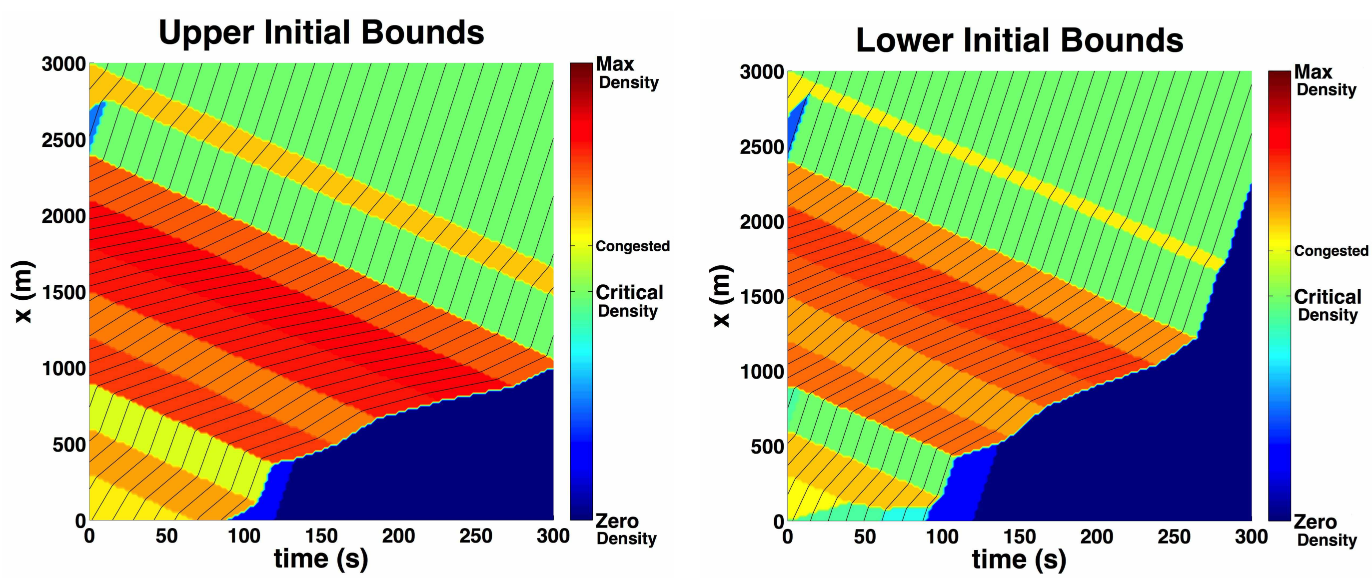

Fig. 3. **Robust control:** Robust control performance in two extreme scenarios associated with an uncertain initial condition. **Left:** Scenario in which the initial condition coefficients are all set at their upper bounds. **Right:** Converse scenario in which all the initial condition coefficients are set to their lower bounds. The robust control outputs in both scenarios present a stable performance (without violating physical limitations or causing unexpected congestion). However, the comparison (very different vehicle density estimations at the final time) shows that under an uncertainty level of 10%, the robust control solution can appear to be quite conservative.

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[4] S. Blandin, D. Work, P. Goatin, B. Piccoli, and A. Bayen, "A general phase transition model for vehicular traffic," *SIAM journal on Applied Mathematics*, vol. 71, no. 1, pp. 107–127, 2011.
[5] X.-Y. Lu, P. P. Varaiya, and R. Horowitz, "An equivalent second order model with application to traffic control," in *Control in Transportation Systems*, 2009, pp. 375–382.
[6] R. Carlson, I. Papamichail, and M. Papageorgiou, "Local feedback-based mainstream traffic flow control on motorways using variable speed limits," *Intelligent Transportation Systems, IEEE Transactions on*, vol. 12, no. 4, pp. 1261–1276, 2011.
[7] M. Papageorgiou, H. Hadj-Salem, and J. Blosseville, "Alinea: A local feedback control law for on-ramp metering," *Transportation Research Record*, no. 1320, 1991.
[8] M. Gugat, M. Herty, A. Klar, and G. Leugering, "Optimal control for traffic flow networks," *Journal of optimization theory and applications*, vol. 126, no. 3, pp. 589–616, 2005.
[9] D. Work, S. Blandin, O. Tossavainen, B. Piccoli, and A. Bayen, "A traffic model for velocity data assimilation," *Applied Mathematics Research eXpress*, vol. 2010, no. 1, pp. 1–35, 2010.
[10] A. Bayen, R. Raffard, and C. Tomlin, "Network congestion alleviation using adjoint hybrid control: Application to highways," *Hybrid Systems: Computation and Control*, pp. 113–129, 2004.
[11] Y. Yuan, J. Van Lint, S. Hoogendoorn, J. Vrancken, and T. Schreiter, "Freeway traffic state estimation using extended kalman filter for first-order traffic model in lagrangian coordinates," in *Networking, Sensing and Control (ICNSC), 2011 IEEE International Conference on*. IEEE, 2011, pp. 121–126.
[12] A. Fügenschuh, S. Göttlich, M. Herty, C. Kirchner, and A. Martin, "Efficient reformulation and solution of a nonlinear pde-controlled flow network model," *Computing*, vol. 85, no. 3, pp. 245–265, 2009.
[13] R. J. Gibbens and F. P. Kelly, "An investigation of proportionally fair ramp metering," in *Intelligent Transportation Systems (ITSC), 2011 14th International IEEE Conference on*. IEEE, 2011, pp. 490–495.
[14] C. Canudas de Wit, D. Jacquet, and D. Koenig, "Optimal ramp metering strategy with extended lwr model, analysis and computational methods," 2005.
[15] C. Canudas de Wit, "Best-effort highway traffic congestion control via variable speed limits," in *Decision and Control and European Control Conference (CDC-ECC), 2011 50th IEEE Conference on*. IEEE, 2011, pp. 5959–5964.
[16] D. Pisarski and C. Canudas de Wit, "Optimal balancing of road traffic density distributions for the cell transmission model," in *Decision and Control (CDC), 2012 IEEE 51st Annual Conference on*. IEEE, 2012, pp. 6969–6974.
[17] D. Bovkovic and M. Krstic, "Backstepping control of chemical tubular reactors," *Computers & chemical engineering*, vol. 26, no. 7, pp. 1077–1085, 2002.
[18] S. Blandin, X. Litrico, and A. Bayen, "Boundary stabilization of the inviscid burgers equation using a lyapunov method," in *Decision and Control (CDC), 2010 49th IEEE Conference on*. IEEE, 2010, pp. 1705–1712.
[19] G. Bastin, B. Haut, J.-M. Coron, and B. d'Andréa Novel, "Lyapunov stability analysis of networks of scalar conservation laws," *Networks and Heterogeneous Media*, vol. 2, no. 4, p. 749, 2007.
[20] A. Fügenschuh, M. Herty, A. Klar, and A. Martin, "Combinatorial and continuous models for the optimization of traffic flows on networks," *SIAM Journal on Optimization*, vol. 16, no. 4, pp. 1155–1176, 2006.
[21] M. Herty and A. Klar, "Simplified dynamics and optimization of large scale traffic networks," *Mathematical Models and Methods in Applied Sciences*, vol. 14, no. 04, pp. 579–601, 2004.
[22] S. Ulbrich, "Optimal control of nonlinear hyperbolic conservation laws with source terms," *Technische Universitaet Muenchen*, 2001.
[23] E. Canepa and C. Claudel, "Exact solutions to traffic density estimation problems involving the lighthill-whitham-richards traffic flow model using mixed integer programming," in *Intelligent Transportation Systems (ITSC), 2012 15th International IEEE Conference on*. IEEE, 2012, pp. 832–839.
[24] K. Moskowitz, "Discussion of freeway level of service as influenced by volume and capacity characteristics by dr drew and cj keese," *Highway Research Record*, vol. 99, pp. 43–44, 1965.
[25] G. Newell, "A simplified theory of kinematic waves in highway traffic, part i,ii and iii," *Transportation Research Part B: Methodological*, vol. 27, no. 4, pp. 281–287, 1993.
[26] G. Dervisoglu, G. Gomes, J. Kwon, R. Horowitz, and P. Varaiya, "Automatic calibration of the fundamental diagram and empirical observations on capacity," in *Transportation Research Board 88th Annual Meeting*, no. 09-3159, 2009.
[27] C. Daganzo, "The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory," *Transportation Research Part B: Methodological*, vol. 28, no. 4, pp. 269–287, 1994.
[28] ——, "A variational formulation of kinematic waves: basic theory and complex boundary conditions," *Transporation Research B*, vol. 39B, no. 2, pp. 187–196, 2005.
[29] ——, "On the variational theory of traffic flow: well-posedness, duality and applications," 2006.
[30] E. Barron and R. Jensen, "Semicontinuous viscosity solutions for hamilton–jacobi equations with convex hamiltonians," *Communications in Partial Differential Equations*, vol. 15, no. 12, pp. 293–309, 1990.
[31] H. Frankowska, "Lower semicontinuous solutions of hamilton-jacobi-bellman equations," *SIAM Journal on Control and Optimization*, vol. 31, no. 1, pp. 257–272, 1993.
[32] M. Crandall and P. Lions, "Viscosity solutions of hamilton-jacobi equations," *Trans. Amer. Math. Soc*, vol. 277, no. 1, pp. 1–42, 1983.
[33] C. Claudel and A. Bayen, "Lax–hopf based incorporation of internal boundary conditions into hamilton-jacobi equation. part ii: Computational methods," *Automatic Control, IEEE Transactions on*, vol. 55, no. 5, pp. 1158–1174, 2010.
[34] ——, "Convex formulations of data assimilation problems for a class of hamilton-jacobi equations," *SIAM Journal on Control and Optimization*, vol. 49, no. 2, pp. 383–402, 2011.
[35] J. Aubin, A. M. Bayen, and P. Saint-Pierre, "Dirichlet problems for some hamilton-jacobi equations with inequality constraints," *SIAM journal on control and optimization*, vol. 47, no. 5, pp. 2348–2380, 2008.
[36] C. Claudel and A. Bayen, "Lax–hopf based incorporation of internal boundary conditions into hamilton–jacobi equation. part i: Theory," *Automatic Control, IEEE Transactions on*, vol. 55, no. 5, pp. 1142–1157, 2010.
[37] P. Mazaré, A. Dehwah, C. Claudel, and A. Bayen, "Analytical and grid-free solutions to the lighthill–whitham–richards traffic flow model," *Transportation Research Part B: Methodological*, vol. 45, no. 10, pp. 1727–1748, 2011.
[38] E. Canepa and C. Claudel, "Spoofing cyber attack detection in probe-based traffic monitoring systems using mixed integer linear programming," in *Computing, Networking and Communications, 2013 International IEEE Conference on*. IEEE, 2013.
[39] M. Hladık, "Interval linear programming: A survey," *Linear Programming-New Frontiers in Theory and Applications*, pp. 85–120, 2010.
[40] C. Claudel, T. Chamoin, and A. Bayen, "Solutions to estimation problems for Hamilton-Jacobi equations using Linear Programming," *To appear, IEEE Transactions on Control Sytems Technology*, 2013.
[41] A. Ben-Tal and A. Nemirovski, "Robust solutions of linear programming problems contaminated with uncertain data," *Mathematical Programming*, vol. 88, no. 3, pp. 411–424, 2000.
[42] V. Gabrel, C. Murat, and N. Remli, "Linear programming with interval right hand sides," *International Transactions in Operational Research*, vol. 17, no. 3, pp. 397–408, 2010.
[43] ——, "Best and worst optimum for linear programs with interval right hand sides," in *Modelling, Computation and Optimization in Information Systems and Management Sciences*. Springer, 2008, pp. 126–134.
[44] M. Minoux, "Robust lp with right-handside uncertainty, duality and applications," 2007.
[45] D. Bertsimas and M. Sim, "The price of robustness," *Operations research*, vol. 52, no. 1, pp. 35–53, 2004.

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